Results and interpretation

Main finding. PR and CKA reveal some first-write timing in controlled transformers. High agreement with a write schedule does not guarantee correct computation; complete algorithm recovery remains unproven.

Methods in brief

We evaluated twelve compiled RASP tasks, nine structural variants, and 36 trained transformers on length-six, eight-token-vocabulary sequences. Compiled controls used 96 calibration and 96 disjoint test sequences. Learned models used 2,029/677/677 training/validation/test sequences and three seeds. Blind predictions were frozen before schedule scoring.

Experiment	Observed result
Numerical policy	A total sample-variance floor of 10^{-12} reduces CKA availability mismatches from 14 to 2 across 54 matched computed-view transitions. Two remaining cases involve helper-state timing.
Sorting diagnosis	Each lane fails 64 of 512 probes; the failure sets are disjoint. Doubling QK-logit scale reduces maximum lane error from 8.52×10^{-4} to 7.15×10^{-7}; decoded accuracy stays 100%.
Rotation	All 36 executed orthogonal transforms preserve 100% output agreement. Maximum posthoc PR/CKA error is 1.42×10^{-13}; fixed original-lane checks change.
Compression	Across twelve tasks, target half-width SVD gives 61.62% token accuracy, random projection 37.01%, and full rank 100%. Attention/MLP hidden dimensions remain fixed.
Blind write timing	On six held-out programs, macro F1 is 0.853 for PR and 0.850 for CKA, versus 0.745 for always predicting a write. A simpler change detector scores 0.876.
PCA interventions	Rank-three counterfactual agreement is 58.85% (113/192 shared pairs across twelve programs), versus 42.19% random and 47.92% self patches. Variable mapping uses labels.
Depth/width controls	Nine crossed branch-depth/branch-count programs give a rank-two realized depth/count design, removing exact collinearity. They do not establish a general complexity law.
Learned models	Across 36 models, task means reach 95.41 to 100% test accuracy; palindrome's majority baseline is already 93.94%. Final learned/compiled CKA is 0.152 to 0.802. Internal circuits remain unknown.

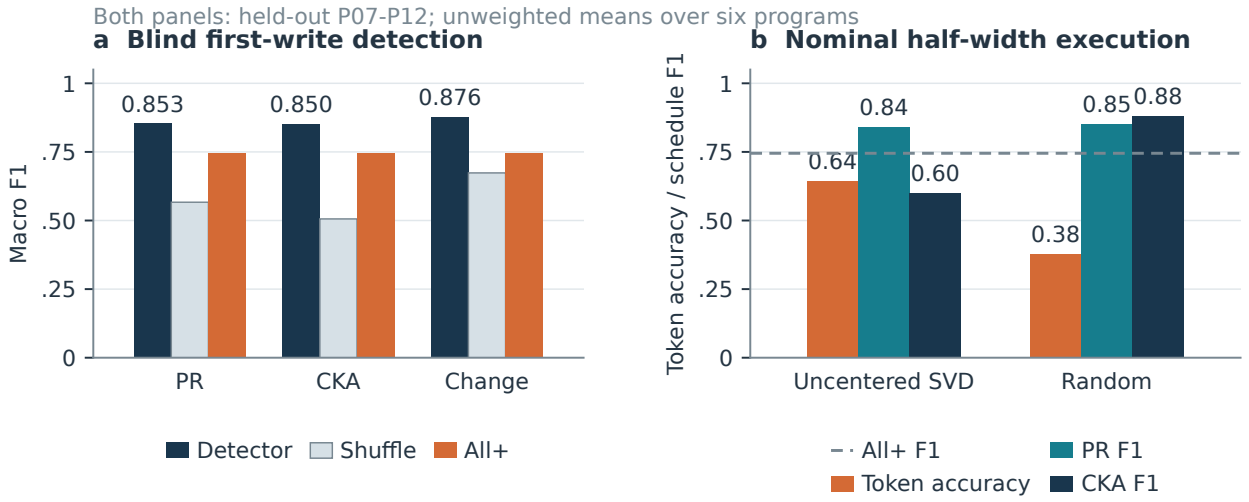


Figure 1. Both panels average the six held-out programs. Random half-width compression scores 0.879 CKA F1 against the original intended schedule while task accuracy is only 37.62%. That schedule is not known causal ground truth for the changed model.

Limits and conclusion

The original fidelity benchmark still uses compiler-known lanes and write times. PR depends on encoding and scale. PCA-to-variable mapping is supervised; 60 undefined counterfactual condition rows remain failures. P11 CKA does not beat its always-write baseline. Erasure, lifetimes, and full circuit recovery are not established. InterpBench provides known circuits through SIIT; our ordinarily trained models do not inherit that ground truth. No direct InterpBench benchmark was run.

Evidence

Recorded verification includes 28 passing tests, five notebooks/37 cells, and 36 checkpoint replays. No scientific experiment was rerun for this brief. RaspFormer code and canonical results, commit e06479c; Tracr, Lindner et al., 2023; CKA, Kornblith et al., 2019; InterpBench, Gupta et al., 2024.

Reference experiments recorded 15 October 2026.